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Phase-field modeling of fission gas bubble growth on grain boundaries and triple junctions in UO_2 nuclear fuel

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ABSTRACT

A phase-field model of fission gas bubble evolution was developed and applied to gain an improved understanding of the microstructure-level processes leading to fission gas release from nuclear fuel, and to inform engineering-scale fission gas release models. The phase-field model accounts for multiple fuel grains and fission gas bubbles and tracks the local concentration of vacancies and gas atoms. The model was used to simulate the growth of grain boundary and triple junction bubbles in a hexagonal periodic 3D grain structure. The fractional coverage of the triple junctions and the number of fully saturated triple junctions (that is, triple junctions fully covered by the gas phase) was calculated, and correlations were developed between these quantities and the grain boundary coverage. The effects of initial triple junction bubble density, vacancy source strength, and bubble semi-dihedral angle on triple junction coverage and saturation were evaluated. High initial triple junction coverage and high bubble semi-dihedral angle can lead to triple junction saturation well before grain boundary percolation. The implications of these findings for engineering-scale fuel performance modeling are discussed.

1. Introduction

The presence of gas bubbles in solids can have an important impact on properties in a wide variety of materials. For example, gaseous porosity frequently forms in metal castings [1–3], reducing the strength of cast parts. Similar gaseous porosity can also form during welding processes [4,5]. In structural metals for nuclear reactors, helium is formed during power operation. This helium precipitates to form bubbles, causing embrittlement of the metals [6]. Within fuels for nuclear reactors, fission results in the formation of a variety of fission products within the fuel matrix. Bubbles of low-solubility fission products, such as the noble gases Xe and Kr, nucleate when gas concentration becomes sufficiently high [7]. The presence of these bubbles impacts the thermal conductivity of the fuel. Additionally, when these bubbles become connected to a free surface of the fuel, the gas is released to the free volume of the fuel-cladding gap and plenum. The release of the fission gases degrades the thermal conductivity of the fuel-cladding gap, leading to higher fuel centerline temperatures. The release of fission gases also increases the gap pressure, potentially accelerating the degradation of the mechanical properties of the cladding that surrounds the fuel. Thus, due to the wide variety of materials that are impacted by the presence of gas bubbles, it is important to develop models for gas bubble evolution at the microstructural level to better understand how such evolution impacts properties and performance.

Within nuclear fuels, the evolution of fission gas is sometimes described as occurring in three stages [8]. In Stage 1, gaseous fission products are produced in the fuel matrix and are transported by diffusion to grain boundaries. In Stage 2, grain boundary and triple junction bubbles nucleate and grow, beginning to interconnect with each other. In Stage 3, a percolated network of gas bubbles connects to a free surface on the fuel pellet, and fission gas is released to the fuel-cladding gap. Some experimental evidence suggests that a percolated network may form along triple junctions prior to percolation of the grain boundary bubbles, creating tunnels along the triple junctions that connect and provide a path for gas venting to the fuel outer surface. An example of such tunnels formed at triple junctions in irradiated, annealed UO_2 is shown in Fig. 1, which also shows that the bubbles on adjacent grain boundaries are not yet percolated. In spite of the Stage 1–3 nomenclature, there may be considerable overlap between the stages. To form of a percolated network of triple junctions, some fraction of the triple junctions must be saturated (fully covered with bubbles). However, to our knowledge it has not been established what fraction of the triple junctions must be saturated for percolation to occur for realistic grain structures, and the relationship between grain

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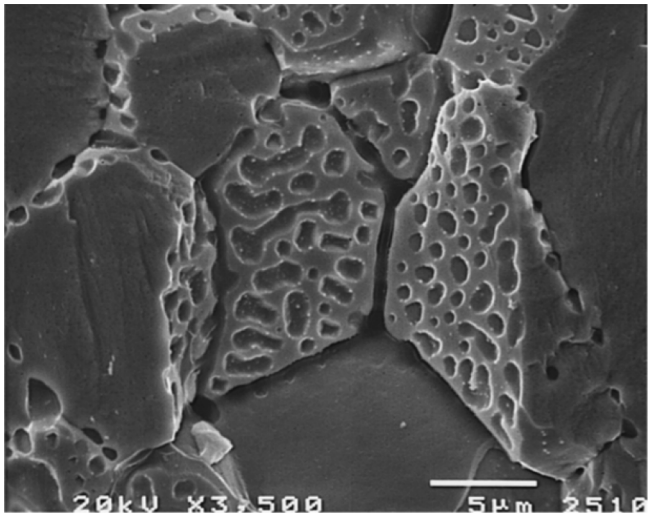


Fig. 1. Scanning electron micrograph of a fracture surface of UO_2 fuel [9], irradiated to 25 GWd/tU and annealed. The three triple junctions that connect near the center of the micrograph are saturated (fully covered by fission gas), forming tunnels. The bubbles on the surrounding grain boundaries are not yet percolated, although some bubbles have begun to interconnect.

boundary coverage and triple junction coverage has not been explored.

Models for the evolution of fission gas bubble microstructure and release processes have been developed in the past using analytical [10–14] and computational [15–21] methods. However, due to the complexity of the microstructures formed during Stage 3, few of these models have attempted to include the effects of triple junction bubbles and the need for a percolated pathway to the surface for fission gas release to occur. In spite of past efforts in simulating fission gas bubble evolution and fission gas release at the microstructural level, some engineering-scale fuel performance codes use an empirical criterion for fission gas release [22,23]: when the grain boundary fractional coverage reaches 0.5 at any position, the gas at that location is released to the fuel-cladding gap, and the need for a percolated pathway to a free surface is not considered in these models.

In this work, we apply a phase-field model to investigate growth of grain boundary and triple junction bubbles and the saturation of triple junctions. The phase-field model explicitly represents the microstructure of multiple fuel grains and the bubble phase, and thus is able to study the simultaneous evolution of grain boundary and triple junction bubbles. The model [24] is a multi-phase, multi-order parameter, multi-component phase model based on the grand-potential formulation originally developed by Plapp [25]. To simulate multiple grains, grain boundary bubbles, and triple junction bubbles simultaneously, 3D simulations are required, so the phase-field model must be efficient. Due to the grand-potential formulation, the model allows the removal of the bulk energy contribution to interfacial energy, allowing increased interface thickness to be used. The increased interface thickness allows the use of a coarser mesh, improving computational efficiency.

The structure of the paper is as follows. In Section 2, the details of the phase-field model and its parameterization are given. In Section 3, the simulation initial conditions and geometry are described, and the results of phase-field simulations are presented. Because some engineering-scale fuel performance codes use grain boundary coverage as the criterion for fission gas release, the grain boundary coverage is calculated, and the triple junction fractional coverage and number of saturated triple junctions is also calculated and plotted as a function of grain boundary coverage. In Section 4, the implications of the phase-field results on engineering-scale fission gas release modeling are discussed. Finally, the results are summarized and final conclusions drawn in Section 5.

2. Phase-field model of fission gas bubble evolution

The microstructure of commercial light water reactor (LWR) fuel consists of multiple grains of UO_2 fuel, intragranular gas bubbles, and intergranular gas bubbles. Because intragranular bubbles do not connect to free surfaces, they do not contribute directly to fission gas release. Their much smaller size would also require a much finer mesh for phase-field simulations. Therefore, for computational efficiency, we do not explicitly include intragranular bubbles in this model.

In the phase-field model, the microstructure is represented with a set of order parameters. For a microstructure consisting of p grains, the individual grains of the fuel matrix are represented by a set of order parameters $\eta_{m1}, \eta_{m2}, \ldots, \eta_{mp}$. Each fission gas bubble is crystallographically indistinguishable from the others, so the bubbles are represented by a single order parameter η_{b0}. Within grain i of the fuel, $\eta_{mi} = 1$, $\eta_{mj} = 0 \; \forall \, j \neq i$, and $\eta_{b0} = 0$. Within the bubble phase, $\eta_{b0} = 1$ and $\eta_{mj} = 0 \; \forall \, j$.

In addition to the local crystal structure, the local concentration of defect species is needed to describe the microstructure. We assume that the dominant defect species are U vacancies and fission gas atoms on U lattice sites. Both U vacancies and interstitials are produced by collision cascades; however, interstitials are much more mobile and therefore diffuse to sinks much more rapidly than vacancies. Therefore, there is a net formation of vacancies within grain interiors which can be represented by a source term for net vacancy production. For fission gas atoms on U lattice sites, the formation of bubbles is driven by the low-solubility Xe and Kr atoms. In this work we focus on the effects of Xe atoms on U sites, since Xe production occurs at a rate nearly ten times that of Kr [7]. The densities of vacancies and gas atoms are represented by variables ρ_v and ρ_g, respectively, with units of number of defects per unit volume. These quantities can be converted to the local composition (mole fraction) c of the U lattice using $c = \rho V_a$, where V_a is the atomic volume occupied by a U atom in the UO_2 crystal structure. V_a was calculated to be $0.0409\,\text{nm}^3$ by dividing the unit cell volume of $0.164\,\text{nm}^3$ (calculated from the lattice constant of 0.547 nm [26]) by 4 U atoms in the unit cell.

2.1. Grand potential functional

To derive the evolution equations for the microstructure, the total grand potential Ω of the system is written as a function of the local grand potential density [24]:

$$\Omega = \int_V \left(m \left[\sum_\alpha \sum_{i=1}^{p_\alpha} \left(\frac{\eta_{\alpha i}^4}{4} - \frac{\eta_{\alpha i}^2}{2} \right) + \sum_\alpha \sum_{i=1}^{p_\alpha} \left(\sum_\beta \sum_{j=1, \alpha i \neq \beta j}^{p_\beta} \frac{\gamma_{\alpha i \beta j}}{2} \eta_{\alpha i}^2 \eta_{\beta j}^2 \right) + \frac{1}{4} \right] + \frac{\kappa}{2} \sum_\alpha \sum_{i=1}^{p_\alpha} \left| \nabla \eta_{\alpha i} \right|^2 + \sum_\alpha h_\alpha \omega_\alpha \right) dV \tag{1}$$

where α and β are indices for phases, i and j index grains of each phase, p_α and p_β are the number of grains of phase α and β, m is a constant free energy barrier coefficient, κ is the gradient energy coefficient (considered to be independent of interface orientation and misinclination here), and the set of parameters $\gamma_{\alpha i \beta j}$ allows the adjustment of interfacial energies between phases and grains. ω_α is the local grand potential density of each phase, and h_α is an switching function for phase α that has value $h_\alpha = 1$ in phase α and $h_\alpha = 0$ in all other phases. The switching function was introduced in Ref. [27] and has the form

$$h_\alpha = \frac{\sum_{i=1}^{p_\alpha} \eta_{\alpha i}^2}{\sum_\beta \sum_{i=1}^{p_\beta} \eta_{\beta i}^2} \tag{2}$$

For the matrix and bubble phases, the switching functions reduce to

$$h_m = \frac{\sum_{i=1}^{p} \eta_{mi}^2}{\eta_{b0}^2 + \sum_{i=1}^{p} \eta_{mi}^2} \tag{3}$$

$$h_b = \frac{\eta_{b0}^2}{\eta_{b0}^2 + \sum_{i=1}^{p} \eta_{mi}^2} \tag{4}$$

The grand potential density for each phase is given by

$$\omega_m = f_m - \mu_g \rho_g - \mu_v \rho_v \tag{5}$$

$$\omega_b = f_b - \mu_g \rho_g - \mu_v \rho_v \tag{6}$$

where f_m and f_b are the Helmholtz free energies of each phase and μ_g and μ_v are the chemical potentials of the gas atoms and vacancies, respectively. The Helmholtz free energies are given by

$$f_m = f_{m,chem} \tag{7}$$

$$f_b = f_{b,chem} \tag{8}$$

where $f_{m,chem}$ and $f_{b,chem}$ are the chemical energy contributions. Due to the excess supply of vacancies relative to gas atoms (discussed further in Section 2.4), we assume that the pressure P within the gas bubbles remains at its equilibrium value as determined by the Laplace-Young equation and therefore there is no mechanical deformation of the fuel matrix near the gas bubbles [7]; thus we neglect any contribution to f_m due to elastic deformation energy.

2.2. Chemical energy contribution and parameterization

The bulk chemical free energy density of the matrix phase, $f_{m,chem}$, is considered first. The solid phase free energy is parameterized using the approach of Ref. [19]. Assuming that the chemical energy of the matrix can be approximated as an ideal solution, the free energy density is written

$$\begin{aligned} f_{m,ideal} = \frac{1}{V_m}\{&RT[c_v \ln c_v + (1-c_v)\ln(1-c_v)] + N_A E_v^f c_v \\ &+ RT[c_g \ln c_g + (1-c_g)\ln(1-c_g)] + N_A E_g^f c_g\} \end{aligned} \tag{9}$$

where V_m is the molar volume, $V_m = V_a N_A$, N_A is Avogadro's number, R is the ideal gas constant, E_v^f is the formation energy of a U vacancy, and E_v^g is the formation (incorporation) energy of a gas (Xe) atom on a U lattice site. As in Ref. [19], we assume $E_v^f = 3$ eV and $E_g^f = 3$ eV, and use $T = 1200$ K as a temperature representative of LWR operation (numerical parameters are summarized in Table 1). To simplify the numerical solution of the governing equations, $f_{m,ideal}$ is approximated with a parabolic function:

$$f_{m,chem} = \frac{1}{2} k_v^m (c_v - c_v^{m,eq})^2 + \frac{1}{2} k_g^m (c_g - c_g^{m,eq})^2 \tag{10}$$

where k_v^m and k_g^m are the curvatures of the parabolas and $c_v^{m,eq}$ and $c_g^{m,eq}$ are the equilibrium composition of vacancies and gas atoms in the UO_2 matrix. The equilibrium compositions are determined from the formation energies and temperature using $c_v^{m,eq} = \exp(-E_v^f/k_B T)$ and $c_g^{m,eq} = \exp(-E_g^f/k_B T)$. The curvatures of the parabolas are set by assuming that at the steady-state vacancy composition during reactor operation, c_v^0, the chemical potential determined from the parabolic approximation is equal to the chemical potential of the ideal solution model. Since $\mu = \frac{\partial f}{\partial \rho} = \frac{\partial f}{\partial c}\frac{\partial c}{\partial \rho} = V_a \frac{\partial f}{\partial c}$,

$$V_a \left.\frac{\partial f_{m,chem}}{\partial c_v}\right|_{c_v^0} = V_a \left.\frac{\partial f_{m,ideal}}{\partial c_v}\right|_{c_v^0} \tag{11}$$

This leads to an expression for k_v^m, assuming $c_v^0 = 0.01$ as in Ref. [19]:

$$\begin{aligned} k_v^m &= \frac{1}{(c_v^0 - c_v^{m,eq})}\left[\frac{RT}{V_m}\left[\ln c_v^0 - \ln\left(1 - c_v^0\right)\right] + \frac{N_A E_v^f}{V_m}\right] \\ &= 4.81 \times 10^{11} \text{J/m}^3 \end{aligned} \tag{12}$$

Table 1
Parameters used for phase-field simulations.

Parameter	Value
T	1200 K
V_a	0.0409 nm^3
E_v^f	3 eV
E_g^g	3 eV
$c_g^{b,eq}$ (intergranular)	0.454
$c_v^{b,eq}$ (intergranular)	0.546
$k_v^m = k_g^m$	4.81×10^{11} J/m^3
$k_v^b = k_g^b$	9.0×10^{10} J/m^3
κ	3.38×10^{-8} J/m
m	3.00×10^{8} J/m^3
γ_{mimj}	1.5
$\theta/2$	50°, 60°, 70°
γ_{mib0}	0.922, 1.5, 11.2
$\dot{F}$	1.09×10^{13} fissions/(cm^3 s)
Y_{Xe}	0.2156
s_g^0	2.35×10^{12} atoms/(cm^3 s)
D_g	0.1 nm^2/s
D_v	0.1 nm^2/s
L	1.56×10^{-11} m^3/(J s)

We also assume $c_g^0 = 0.01$ as in Ref. [19], and thus $k_g^m = k_v^m$.

The bulk chemical free energy density of the gas bubble phase, $f_{b,chem}$, is considered next. The bubble phase is considered to be a mixture of vacancies and gas atoms with no U atoms present. The bubble phase can be treated as a van der Waals gas [7], in which the gas atoms are assumed to have a hard-sphere exclusion volume characterized by the parameter b. For Xe, $b = 0.085$ nm^3/atom [7]. (Due to the high density of Xe atoms in bubbles, the long-range attractive interactions in the van der Waals gas are neglected.) The chemical free energy density of the bubble phase is given by the Helmholtz free energy density of a van der Waals gas [28]:

$$f_{b,vdW} = n_g kT\left[\ln\left(\frac{1}{n_Q\left(\frac{1}{n_g} - b\right)}\right) - 1\right] + f_0 \tag{13}$$

where n_g is the number density of gas atoms, $n_Q = \left(\frac{mk_BT}{2\pi\hbar}\right)^{3/2}$ is the quantum concentration, m is the mass of a Xe atom, and f_0 is an offset between the solid and gas bubble phase free energies. n_g can be put in terms of model variables using $n_g = c_g n_U$ (where $n_U = 1/V_a$ is the number density of U atoms in the UO_2 lattice) as long as $c_v + c_g = 1$ holds. f_0 is determined by setting the minima of the matrix and gas bubble phase chemical free energies to be equal, so that when the matrix and bubble phases are in equilibrium at compositions along the tie line between the minima, the matrix phase vacancy and gas concentrations will be at the expected values $c_v^{m,eq}$ and $c_g^{m,eq}$ determined by the defect formation energies. This results in $f_0 = 2.22 \times 10^9$ J/m^3.

The free energy of the gas bubble phase can be more easily understood by considering the Gibbs triangle representing the composition of U lattice sites in Fig. 2. The corners of the triangle represent 100% U atoms, vacanices, and Xe atoms, and are labeled accordingly. Along the edge linking the Vac and Xe corners, no U is present, so $c_v + c_g = 1$. Thus, along that edge, the Helmholtz free energy of the gas bubble phase is given by Eq. (13). A plot of Eq. (13) along that edge is also shown in Fig. 2.

As seen in Fig. 2, $f_{b,vdW}$ increases dramatically when $c_g > 0.49$. This occurs as $\frac{1}{n_g} - b$ approaches 0, and corresponds physically to the density at which the volume occupied by a Xe atom approaches the hard sphere exclusion volume b. By setting $\frac{1}{n_g} - b = \frac{1}{c_g n_U} - b = 0$, it can be determined that $f_{b,vdW} \to \infty$ as $c_g \to 0.494$. To simplify numerical calculations, for phase-field simulations a parabolic approximation was fit

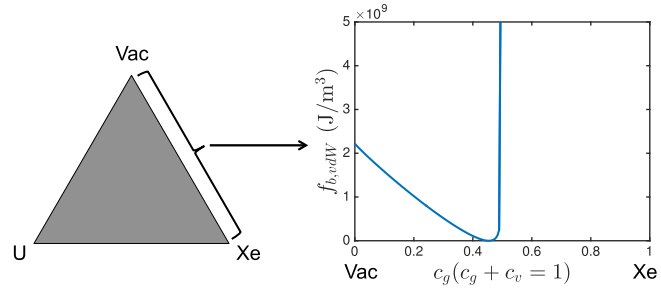


Fig. 2. Left, the Gibbs triangle representation of composition of the U lattice sites. The corners correspond to 100% U atoms, vacancies, and Xe atoms. Along the Vac-Xe edge, no U atoms are present, so $c_g + c_v = 1$. Right, the Helmholtz free energy of the gas bubble phase is plotted along the Vac-Xe edge using Eq. (13), with $c_g = 0$ corresponding to the pure vacancy corner and $c_g = 1$ corresponding to the pure Xe corner.

to the Helmholtz free energy:

$$f_{b,chem} = \frac{1}{2}k_v^b(c_v - c_v^{b,eq})^2 + \frac{1}{2}k_g^b(c_g - c_g^{b,eq})^2 \tag{14}$$

The minimum of the parabolic free energy was set to occur at the minimum of the van der Waals free energy, resulting in $c_g^{b,eq} = 0.454$, $c_v^{b,eq} = 0.546$. This composition is found along the Vac-Xe edge of the Gibbs triangle in Fig. 2. From the data of Ref. [29], at $c_g = 0.454$, bubble pressure $P = 465$ MPa. This is in the range of the experimentally measured intergranular gas bubble pressures reported in Ref. [30]. Thus, the initial composition of the bubbles in simulations is set at $c_g^{b,eq} = 0.454$ and $c_v^{b,eq} = 0.546$. Because composition in the gas bubbles will generally not deviate far from the minimum of the free energy for intergranular bubbles, k_v^b and k_g^b were set by assuming $k_v^b = k_g^b$ and fitting $f_{b,chem}$ to $f_{b,vdW}$ in the range $0.42 < c_g < 0.49$, resulting in $k_v^b = k_g^b = 9.0 \times 10^{10}$ J/m^3.

2.3. Interfacial energy and parameterization

In Ref. [31], the grain boundary energies for UO_2 were calculated for a variety of grain boundary types and misorientations. Based on these results, the grain boundary energy is approximately $\sigma_{mm} = 1.5$ J/m^2. The semi-dihedral angle of a grain boundary bubble is determined by the balance of grain boundary energy and bubble-matrix interfacial energy using Young's equation. Although White reported the semi-dihedral angle of intergranular bubbles in UO_2 to be approximately 50° [14], others have observed significant local variations in the semi-dihedral angle within a highly localized area of the fuel [32]. We therefore consider 50° as a baseline value for the semi-dihedral angle, and vary it to study its effect on triple junction saturation. The interfacial energies are varied to consider semi-dihedral angles of 50°, 60°, and 70°.

Using the grand potential functional of Eq. (1), the chemical energy does not contribute to the interfacial energy between phases [24]. The interfacial energy $\sigma_{\alpha i\beta j}$ between grain i of phase α and grain j of phase β is given by [24,27,33]

$$\sigma_{\alpha i\beta j} = g(\gamma_{\alpha i\beta j})\sqrt{\kappa m} \tag{15}$$

where $g(\gamma_{\alpha i\beta j})$ is a dimensionless function of $\gamma_{\alpha i\beta j}$ that in general must be evaluated numerically. However, for the special case $\gamma_{\alpha i\beta j} = 1.5$, $g(\gamma_{\alpha i\beta j} = 1.5) = \sqrt{2}/3$. Here, we choose the interface between any grain i of the matrix phase and grain j of the matrix phase to have $\gamma_{mimj} = 1.5$. For this special case, analytical expressions can be used that relate κ and m to the interfacial energy and the width of the interface l_{int} [27,33]:

$$\kappa = \frac{3}{4}\sigma_{mm}l_{int} \tag{16}$$

$$m = \frac{6\sigma_{mm}}{l_{int}} \tag{17}$$

To resolve the intergranular bubbles, l_{int} is chosen to be 30 nm, and $\sigma_{mm} = 1.5$ J/m^2, resulting in $\kappa = 3.38 \times 10^{-8}$ J/m and $m = 3.00 \times 10^8$ J/m^3. To control the bubble-matrix interfacial energy, since κ and m are fixed, the parameters γ_{mib0} must be determined to obtain the correct value of σ_{mb} to obtain semi-dihedral angles 50°, 60°, and 70°. To determine the value of γ_{mib0}, using Eq. (15),

$$\sigma_{mb} = g(\gamma_{mib0})\sqrt{\kappa m} \tag{18}$$

$$\sigma_{mm} = g(\gamma_{mimj})\sqrt{\kappa m} \tag{19}$$

Dividing Eq. (18) by Eq. (19),

$$\frac{\sigma_{mb}}{\sigma_{mm}} = \frac{g(\gamma_{mib0})}{g(\gamma_{mimj})} \tag{20}$$

Rearranging Eq. (20),

$$g(\gamma_{mib0}) = g\left(\gamma_{mimj} = 1.5\right)\frac{\sigma_{mb}}{\sigma_{mm}} = \frac{\sqrt{2}}{3}\frac{\sigma_{mb}}{\sigma_{mm}} \tag{21}$$

A polynomial approximation has been fit to numerical results that allows γ to be found as a function of g [34]:

$$\gamma = (-5.288g^8 - 0.09364g^6 + 9.965g^4 - 8.813g^2 + 2.007)^{-1} \tag{22}$$

Using this approximation, the appropriate value of γ_{mib0} can be determined which results in semi-dihedral angles 50°, 60°, and 70°. These values are summarized in Table 1.

2.4. Evolution equations

From the grand potential functional of Eq. (1), the Allen–Cahn equations for evolution of the order parameters can be derived:

$$\frac{\partial\eta_{\alpha i}}{\partial t} = -L\frac{\delta\Omega}{\delta\eta_{\alpha i}}$$
$$\frac{\partial\eta_{\alpha i}}{\partial t} = -L\left[m\left(\eta_{\alpha i}^3 - \eta_{\alpha i} + 2\eta_{\alpha i}\sum_{\beta}\sum_{j=1,\alpha i\neq\beta j}^{p_\beta}\gamma_{\alpha i\beta j}\eta_{\beta j}^2\right) - \kappa\nabla^2\eta_{\alpha i} + \sum_{\alpha}\frac{\partial h_\alpha}{\partial\eta_{\alpha i}}\omega_\alpha\right] \tag{23}$$

where $\frac{\delta\Omega}{\delta\eta_{\alpha i}}$ is the variational derivative of Ω with respect to order parameter $\eta_{\alpha i}$ and L is the order parameter mobility, which in general is a function of order parameters and concentration. Parameterization of L will be discussed later in this section.

To evolve the gas and vacancy concentrations, it is more convenient to write the evolution equations in terms of chemical potentials μ_v and μ_g and use the chemical potentials as the field variables rather than compositions or densities. To enable this, the grand potentials in each phase, Eqs. (5) and (6), must be expressed in terms of μ_v and μ_g for use in Eq. (23). For the matrix phase, the relation $\mu_g = V_a\frac{\partial f_{m,chem}}{\partial c_g} = V_a k_g^m\left(c_g - c_g^{m,eq}\right)$ can be re-arranged to yield

$$c_g = \frac{\mu_g}{V_a k_g^m} + c_g^{m,eq} \tag{24}$$

Similarly,

$$c_v = \frac{\mu_v}{V_a k_v^m} + c_v^{m,eq} \tag{25}$$

Eqs. (24) and (25) can be used in conjunction with $\rho_g = \frac{c_g}{V_a}$, $\rho_v = \frac{c_v}{V_a}$, Eqs. (5), (7), and (10) to obtain the grand potential density as a function of chemical potentials:

$$\omega_m = -\frac{1}{2}\frac{\mu_v^2}{V_a^2 k_v^m} - \frac{\mu_v}{V_a}c_v^{m,eq} - \frac{1}{2}\frac{\mu_g^2}{V_a^2 k_g^m} - \frac{\mu_g}{V_a}c_g^{m,eq} \tag{26}$$

Similarly, the grand potential density of the bubble phase can be

expressed as

$$\omega_b = -\frac{1}{2}\frac{\mu_v^2}{V_a^2 k_v^b} - \frac{\mu_v}{V_a}c_v^{b,eq} - \frac{1}{2}\frac{\mu_g^2}{V_a^2 k_g^b} - \frac{\mu_g}{V_a}c_g^{b,eq} \tag{27}$$

The evolution equations for μ_g and μ_v are

$$\frac{\partial \mu_g}{\partial t} = \frac{1}{\chi_g}\left[\nabla\cdot(D_g\chi_g\nabla\mu_g) + s_g - \sum_\alpha \sum_{i=1}^{P_\alpha} \frac{\partial \rho_g}{\partial \eta_{\alpha i}}\frac{\partial \eta_{\alpha i}}{\partial t}\right] \tag{28}$$

$$\frac{\partial \mu_v}{\partial t} = \frac{1}{\chi_v}\left[\nabla\cdot(D_v\chi_v\nabla\mu_v) + s_v - \sum_\alpha \sum_{i=1}^{P_\alpha} \frac{\partial \rho_v}{\partial \eta_{\alpha i}}\frac{\partial \eta_{\alpha i}}{\partial t}\right] \tag{29}$$

where χ_g and χ_v are the susceptibilities (discussed further below), D_g and D_v are the diffusion coefficients, and s_g and s_v are the source terms for production of Xe atoms and U site vacancies.

The source term $s_g = s_g^0 h_m$ is given by a constant rate of Xe production, s_g^0, times the switching function h_m, which has a value of 1 in the fuel matrix and 0 inside the bubble. This is to limit production of new Xe atoms to the fuel matrix in the model. The Xe production rate $s_g^0 = \dot{F} Y_{Xe}$, where $\dot{F}$ is the fission rate density and Y_{Xe} is the fission yield of Xe. $\dot{F}$ is estimated to be 1.09×10^{13} fissions/(cm^3 s) based on typical operating values for a LWR [7]. Y_{Xe} is taken to be 0.2156 based on the thermal neutron Xe yield for U-235 [35]. The net vacancy production rate is similarly given by $s_v = s_v^0 h_m$. s_v^0 has not been determined to our knowledge, but it is generally believed that under normal operating conditions, there is an adequate supply of vacancies to maintain gas bubbles in equilibrium as determined by the Laplace-Young equation [7]. Therefore, s_v^0 is varied parametrically in the range $5s_g^0$ to $20s_g^0$, and the effect of this parameter on grain boundary coverage and triple junction coverage is studied.

The susceptibility χ describes the relationship between solute density and its chemical potential: $\chi = \frac{\partial \rho}{\partial \mu}$ [25]. This relationship differs based on the phase of the system, so χ is interpolated based on the local phase using the switching functions h_α. For gas atoms,

$$\chi_g = h_m\chi_g^m + h_b\chi_g^b \tag{30}$$

where $\chi_g^m = \frac{\partial \rho_g^m}{\partial \mu_g}$ and $\chi_g^b = \frac{\partial \rho_g^b}{\partial \mu_g}$. Since the governing equations are in terms of the chemical potentials, the susceptibilities must also be expressed in terms of chemical potentials, which can be done as follows. Using $\rho_g^m = \frac{c_g}{V_a}$, and substituting for c_g using Eq. (24), $\chi_g^m = \frac{\partial \rho_g^m}{\partial \mu_g} = \frac{1}{V_a^2 k_g^m}$. Similar expressions can be derived for χ_v^m, χ_g^b, and χ_v^b, resulting in

$$\chi_g = h_m\frac{1}{V_a^2 k_g^m} + h_b\frac{1}{V_a^2 k_g^b} \tag{31}$$

$$\chi_v = h_m\frac{1}{V_a^2 k_v^m} + h_b\frac{1}{V_a^2 k_v^b} \tag{32}$$

The diffusion coefficient for Xe, D_g, was set to 0.1 nm^2/s based on experimental measurement in stoichiometric UO_2 [36]. Although the diffusivity of Xe in the gas phase should be expected to be much higher, microstructural evolution is expected to be dominated by transport through the solid phase, where gas atoms are generated. Therefore, for simplicity, we set the diffusion coefficient to be the same in both phases. Additionally, for simplicity the diffusion coefficient for U vacancies, D_v, was set to the same value.

The governing equations are non-dimensionalized using length scale $l^* = 1$ nm, time scale $\tau^* = 0.1$ s, and energy density scale $E^* = 64 \times 10^9$ J/m^3. Finally, we describe the parameterization of the order parameter mobility L. In the simulation configuration used, the grain boundaries between fuel matrix grains are static, and the change in microstructure is only due to the motion of matrix-bubble interfaces. We therefore use a constant L for all interfaces, and set L throughout based on the matrix-bubble interfaces. We assume that the motion of the bubble-matrix interface is limited by diffusion, and thus set the non-dimensionalized order parameter mobility $\bar{L}$ high enough that interface motion was diffusion-limited. $\bar{L} = 0.1$ was chosen based on the fact that further increases did not result in changes to the microstructure. To convert this to a dimensional value, since the dimensions of L are volume/(energy × time), $L = \frac{\bar{L}}{E^*\tau^*}$. This is equivalent to a dimensional value $L = 1.56 \times 10^{-11}$ m^3/(J s).

The non-dimensionalized governing equations were discretized using the MOOSE framework [37]. Hexahedral 3D mesh elements with linear Lagrange shape functions were used for spatial discretization. Mesh adaptivity was used, with three levels of refinement and a minimum element size of $\Delta x = \Delta y = \Delta z = 10$ nm so that the element size in the interfacial regions is 1/3 of the interfacial width (l_{int} = 30 nm). Periodic boundary conditions were used in all directions. The second-order backward differentiation formula was used for time integration, and adaptive time stepping was used with the MOOSE IterationAdaptiveDT time stepper, with eight optimal nonlinear iterations and an iteration window of two.

3. Phase-field simulation results

3.1. Initial conditions

To simulate the growth of triple junction bubbles, a 3D domain with multiple grains is required. However, this must be balanced against the need for a simulation domain small enough to be computationally tractable. To balance these requirements, a periodic structure of four uniform hexagonal grains in the x-y plane is extended in the z-direction, as shown in Fig. 3. The grains are represented by order parameters η_{m1} through η_{m4} and the bubbles are represented by η_{b0}. This structure has eight triple junctions parallel to the z-direction. Because all grain boundaries are flat, this configuration is metastable with respect to grain growth, allowing the evolution of fission gas bubble microstructure to be studied independently of grain boundary motion.

The hexagons in the grain structure in these simulations have an edge-to-edge distance d of 0.6 μm. This is smaller than the typical grain size of ~ 5 μm in UO_2 in commercial LWRs. However, assuming that the density of grain boundary bubbles per unit grain boundary area, N_{GB}, is approximately 20/ μm^2 at the early stages of bubble formation [14], the

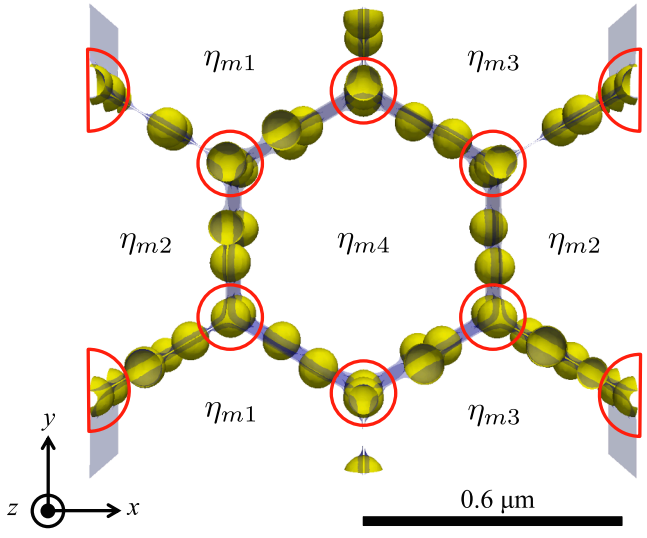


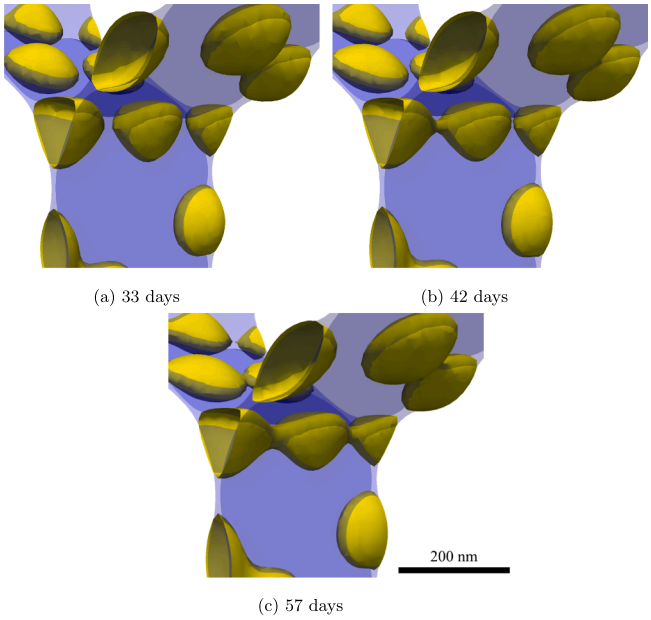
Fig. 3. Initial conditions for phase-field simulations. Intergranular fission gas bubbles are yellow, and grain boundaries are in transparent blue-grey. The order parameter representing each grain is indicated, and periodic boundary conditions are used in all directions. Triple junctions are circled in red, with the triple junctions at the left and right edges of the domain in half-circles to indicate that half of each triple junction is across the periodic boundary in the x-direction. Eight total triple junctions are present.

Table 2
Geometries used in phase-field simulations.

N_{TJ} (μm^{-1})	l_{TJ} (μm)	L_z (μm)	A_{GB} (μm^2)	n_{GB}
4.46	0.224	0.448	1.86	37
6.25	0.16	0.32	1.33	27
7.14	0.14	0.28	1.16	23
8.93	0.112	0.224	0.931	19

characteristic spacing between grain boundary bubbles is $l_{GB} = 1/\sqrt{N_{GB}} = 0.224$ μm, smaller than the grain size. This configuration is therefore large enough to allow a reasonable number of bubbles to be placed on the grain boundaries and triple junctions, and for the effect of interactions between triple junction bubbles and nearby grain boundary bubbles to be studied. The size of the simulation domain in the x, y, and z directions is defined as L_x, L_y, and L_z, respectively. $L_x = 2d = 1.2$ μm for the hexagonal geometry. It can be shown that for the uniform hexagons, $L_y = \sqrt{3}L_x/2 = 1.04$ μm. L_z was varied depending on the density of triple junction bubbles per unit triple junction length, N_{TJ}, as described further below. The grain boundary area for this configuration without bubbles is $A_{GB} = 2\sqrt{3}L_xL_z$.

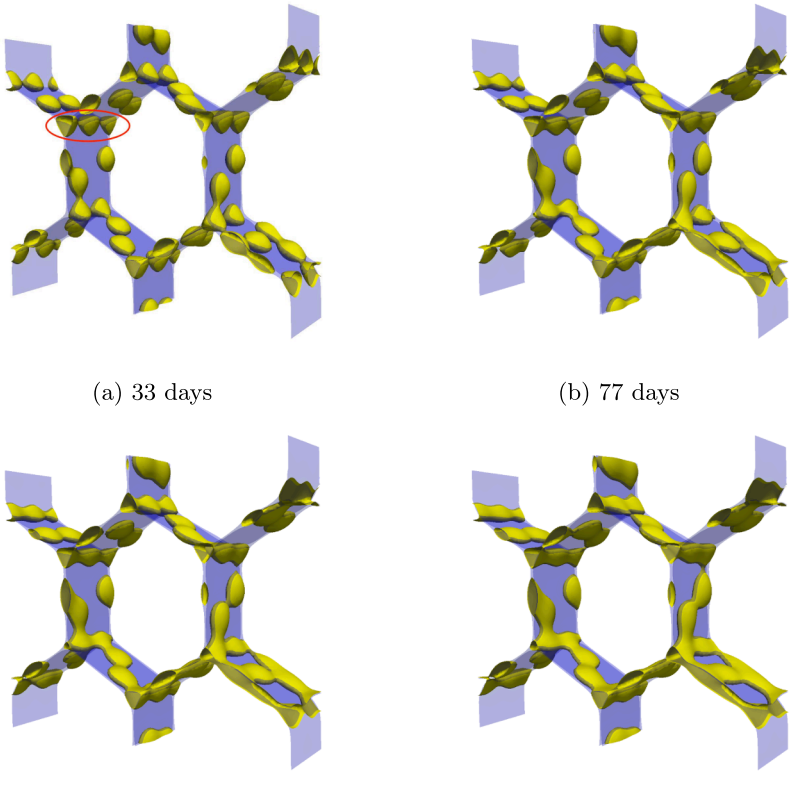
Grain boundary bubbles were placed randomly on the grain boundaries with density $N_{GB} = 20/\mu m^2$, based on experimental observations [14]. To our knowledge, no experimental data is available to set N_{TJ} in the initial conditions. It is expected that triple junctions are more energetically favorable sites for nucleation than grain boundaries [38]. This would lead to the characteristic spacing between triple junction bubbles, $l_{TJ} = 1/N_{TJ}$, being smaller than l_{GB} under conditions of one-off nucleation, which was determined to be a good assumption [14]. However, the nucleation rate is a very sensitive function of parameters that are not well known [38]. Therefore, we did not attempt to calculate the relative nucleation rates of grain boundary and triple junction bubbles, and instead investigated the effect of varying l_{TJ}, assuming that $l_{TJ} \leqslant l_{GB}$. This was done by placing two bubbles on each triple junction in the initial conditions, and varying L_z to obtain the desired l_{TJ}, while maintaining $N_{GB} = 20/\mu m^2$ by varying the number of grain boundary bubbles, n_{GB}. The geometries investigated are



(a) 33 days (b) 42 days

(c) 57 days

Fig. 5. Closeup of the triple junction circled in red in Fig. 4a, showing details of the interconnection process.



(a) 33 days (b) 77 days

(c) 106 days (d) 145 days

Fig. 4. Evolution of microstructure with $s_v^0 = 10s_g^0$, and $L_z = 0.28$ μm, corresponding to $N_{TJ} = 7.14/\mu m$, and a semi-dihedral angle $\theta/2 = 50°$. The grain boundaries are shown in semitransparent blue and the gas bubbles are shown in yellow. The triple junction circled in red in (a) is shown in detail in Fig. 5.

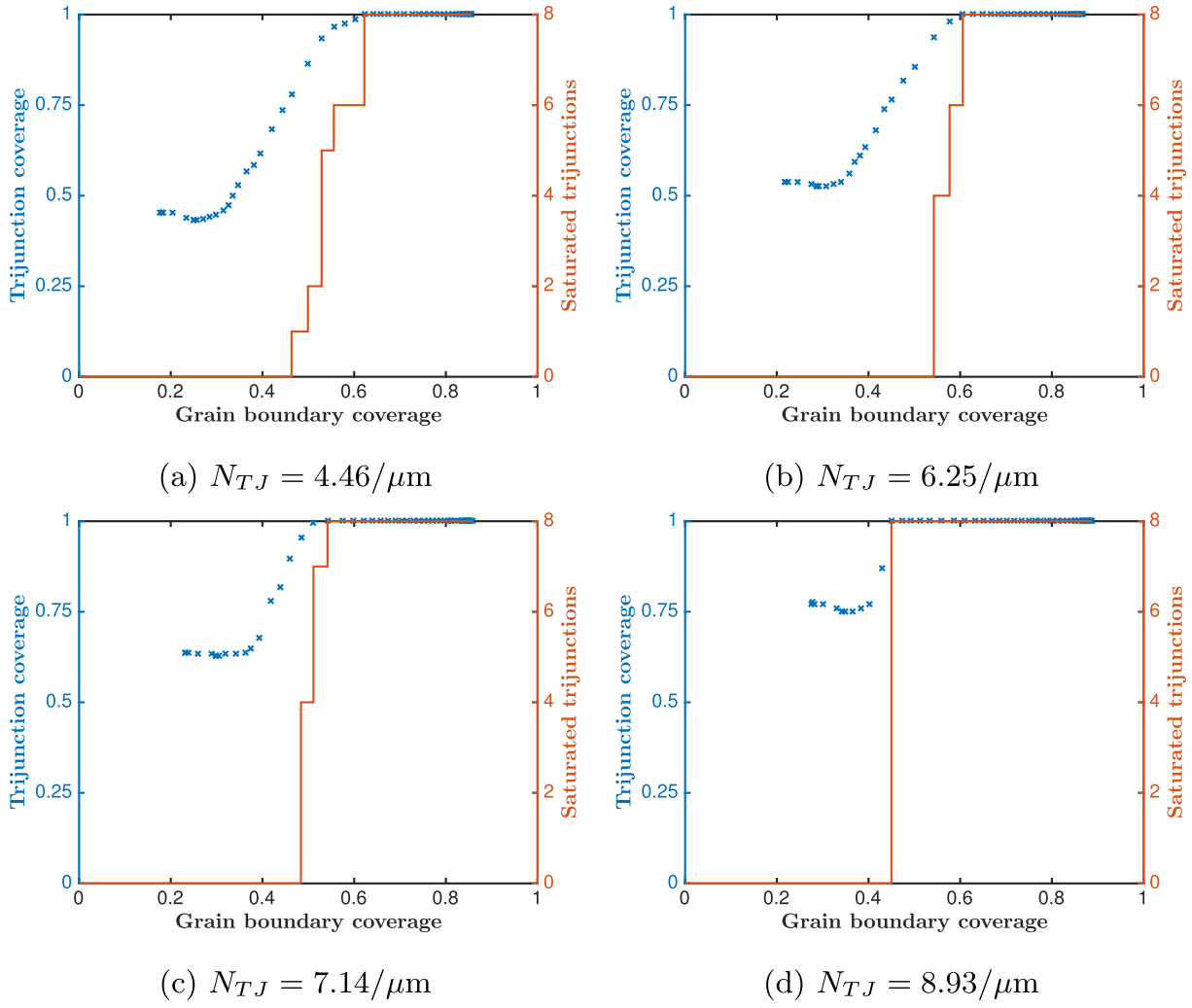


Fig. 6. Evolution of fractional triple junction coverage (X_{TJ}) and number of saturated triple junctions (n_s) as a function of grain boundary coverage by fission gas bubbles, for varying linear density of bubbles on triple junctions N_{TJ}. As N_{TJ} increases, the initial triple junction coverage increases, and the individual triple junctions saturate sooner with respect to grain boundary coverage. The data of Fig. 6c corresponds to the microstructural images shown in Fig. 4.

summarized in Table 2. The bubbles in the initial conditions are spherical with a radius of 44 nm [16], and rapidly equilibrate to their expected shape (lenticular for grain boundary bubbles and elongated triangular for triple junction bubbles) in the early stages of each simulation.

3.2. Microstructural evolution

The evolution of the microstructure for a simulation with $N_{TJ} = 7.14/\,\mu m$, $s_v^0 = 10 s_g^0$, and semi-dihedral angle 50° is shown in Fig. 4. In keeping with assumption that nucleation at triple junctions is more likely than at grain boundaries, this value of N_{TJ} corresponds to $l_{TJ} = 0.14\,\mu m$, smaller than $l_{GB} = 0.224\,\mu m$. Grain boundaries are shown in semitransparent blue, and fission gas bubbles are shown in yellow. In the early stages of evolution, grain boundary bubbles have begun to take the expected lenticular shape, as seen in Fig. 4a. The shape of triple junction bubbles is approximately triangular in cross-section. A few bubbles have begun to coalesce. As time progresses, bubbles grow and more coalescence occurs, see Fig. 4b–c. The grain boundaries remain in their initial metastable configuration, as expected. The triple junction bubbles begin to elongate along the z-direction and connect with each other to form a saturated triple junction. A detailed view of the triple junction circled in red in Fig. 4a is shown in Fig. 5. Fig. 5c shows the fully saturated triple junction. In Fig. 4d, the microstructure is shown at 145 days, the time when all eight triple junctions become fully covered by bubbles. At this time, the grain boundaries are 54% covered by bubbles.

3.3. Grain boundary coverage and triple junction coverage

The grain boundary coverage and the triple junction coverage were calculated as a function of time from the phase-field simulations. The grain boundary coverage X_{GB} is defined as the area covered by bubbles on the grain boundaries, A_b, divided by the grain boundary area of the hexagonal grain configuration in the absence of bubbles, A_{GB}. The triple junction coverage X_{TJ} is defined as the length covered by bubbles along the triple junctions, L_b, divided by the total length of the triple junctions, which for this configuration is equal to $8L_z$. Each position is considered to be covered by a bubble if $\eta_{b0} > 0.5$ there. In addition, the number of triple junctions that are fully saturated, n_s, is calculated as a function of time. A triple junction is considered to be saturated when $\eta_{b0} > 0.5$ along its entire length.

Currently, some engineering scale fuel performance codes use $X_{GB} = 0.5$ in each volume element as the sole criterion for fission gas release to the fuel pin free volume. This does not account for the need for a percolated pathway to a free surface. To gain insight on whether this assumption is valid, it is useful to plot X_{TJ} and n_s versus X_{GB}. Such plots are shown in Fig. 6 for varying N_{TJ} with vacancy source strength

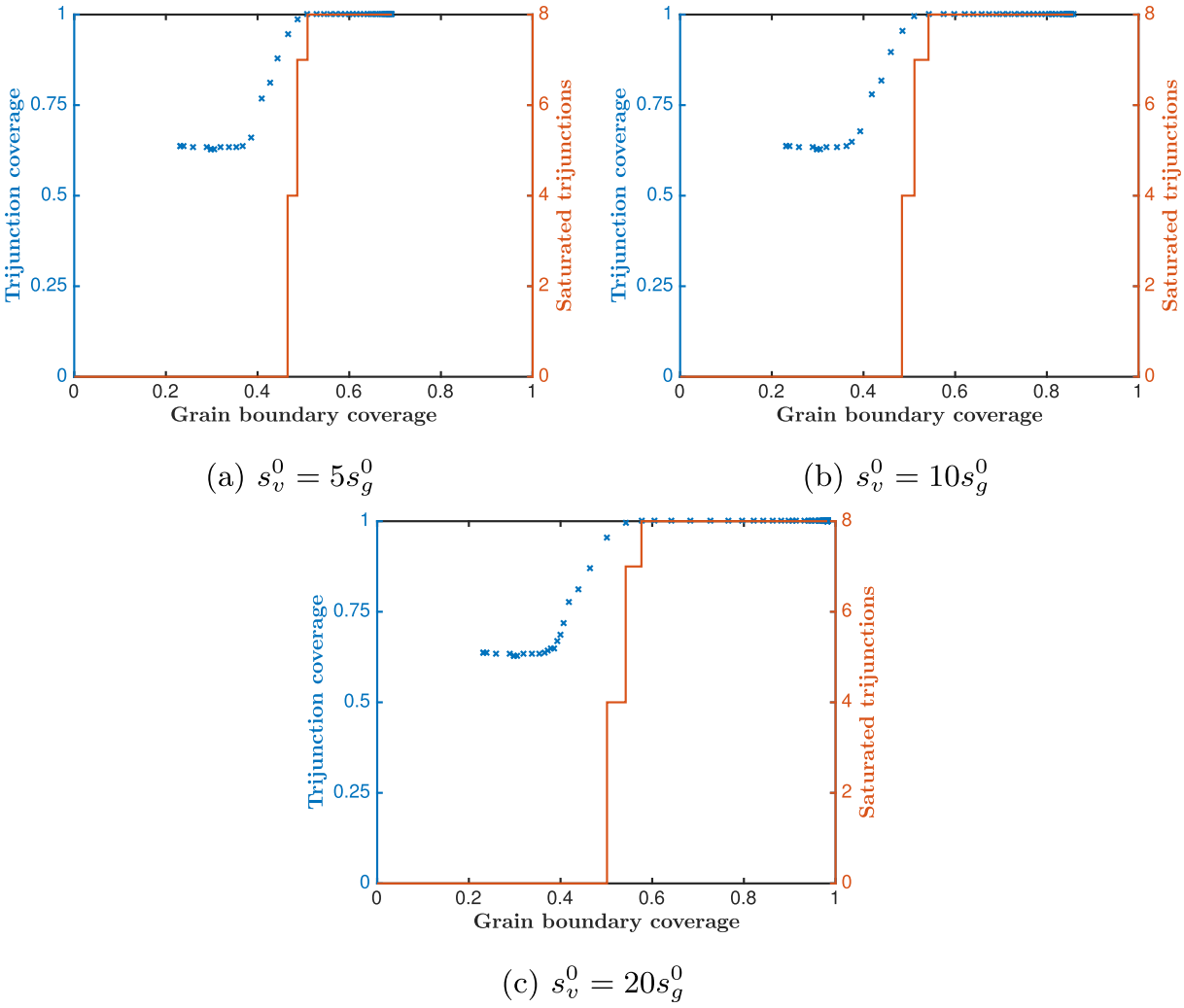


Fig. 7. Evolution of fractional triple junction coverage (X_{TJ}) and number of saturated triple junctions (n_s) as a function of grain boundary coverage by fission gas bubbles, for varying vacancy source strength s_v^0. As s_v^0 increases, the triple junctions become saturated later with respect to grain boundary coverage (both curves shift to the right). For these conditions, semi-dihedral angle = 50° and N_{TJ} = 7.14/μm.

of $s_v^0 = 10s_g^0$ and semi-dihedral angle 50°.

Several trends are apparent in comparing Figs. 6a–d. The initial value of X_{TJ} increases with increasing N_{TJ} since more of the triple junctions are covered by the initial bubbles. The initial value of X_{GB} also increases slightly since the number of triple junction bubbles remains constant, and the triple junction bubbles cover a small amount of the grain boundaries. However, this effect is small. During the initial equilibration phase of the bubbles, X_{GB} increases rapidly while X_{TJ} remains relatively constant, leading to a relatively flat portion of the curve at the beginning. This portion of the curve corresponds to the bubbles rapidly taking their equilibrium shape, during which time the grain boundary bubbles spread more than the triple junction bubbles. After this equilibration period, X_{TJ} begins to increase. The rate of increase in X_{TJ} is higher with increasing N_{TJ}. The triple junctions become fully saturated for each of the configurations in Fig. 6. As expected, for higher initial N_{TJ}, the triple junctions become saturated sooner with respect to X_{GB}; at an initial N_{TJ} of 4.46/μm, the triple junctions become fully saturated at X_{GB} = 0.62, whereas at an initial N_{TJ} of 8.93/μm, the triple junctions become fully saturated at X_{GB} = 0.45.

The progress of the total number of saturated triple junctions, n_s, is also shown in Fig. 6. At lower N_{TJ}, saturation happens later and over a greater range of X_{GB}, whereas for larger N_{TJ}, saturation happens earlier and over a smaller range.

Because the strength of the vacancy source term s_v^0 is not known, a set of simulations was run with varying s_v^0 to assess its effect. The configuration with N_{TJ} = 7.14/μm and semi-dihedral angle 50° was used, and source strengths $s_v^0 = 5s_g^0$ and $s_v^0 = 20s_g^0$ were simulated in addition to the previously discussed $s_v^0 = 10s_g^0$. Results of these simulations are shown in Fig. 7. Comparing these figures, it can be observed that as s_v^0 increases, the triple junctions become fully saturated slightly later with respect to grain boundary coverage. When $s_v^0 = 5s_g^0$, all triple junctions become saturated at X_{GB} = 0.51, whereas when $s_v^0 = 20s_g^0$, all triple junctions become saturated when X_{GB} = 0.58. Considering the large change in the vacancy source term between these cases, the value of X_{GB} at full triple junction saturation does not change much. Thus, the correlation between grain boundary coverage and triple junction coverage is relatively insensitive to s_v^0, which is fortunate because its value is not well known. (It should be noted that plots of X_{GB} and X_{TJ} versus time are more sensitive to the value of s_v^0; but the correlation between the two parameters is much less sensitive.).

The effect of different semi-dihedral angles was also considered in simulations, with the results shown in Fig. 8. Semi-dihedral angles of $\theta/2$ = 60° and 70° were considered in addition to the previously discussed 50°. Fig. 8 shows that as $\theta/2$ increases, the triple junctions begin to saturate earlier with respect to grain boundary coverage, starting at X_{GB} = 0.34. This is because the grain boundary bubbles with greater semi-dihedral angles cover a smaller fraction of the grain boundaries for equal bubble volume.

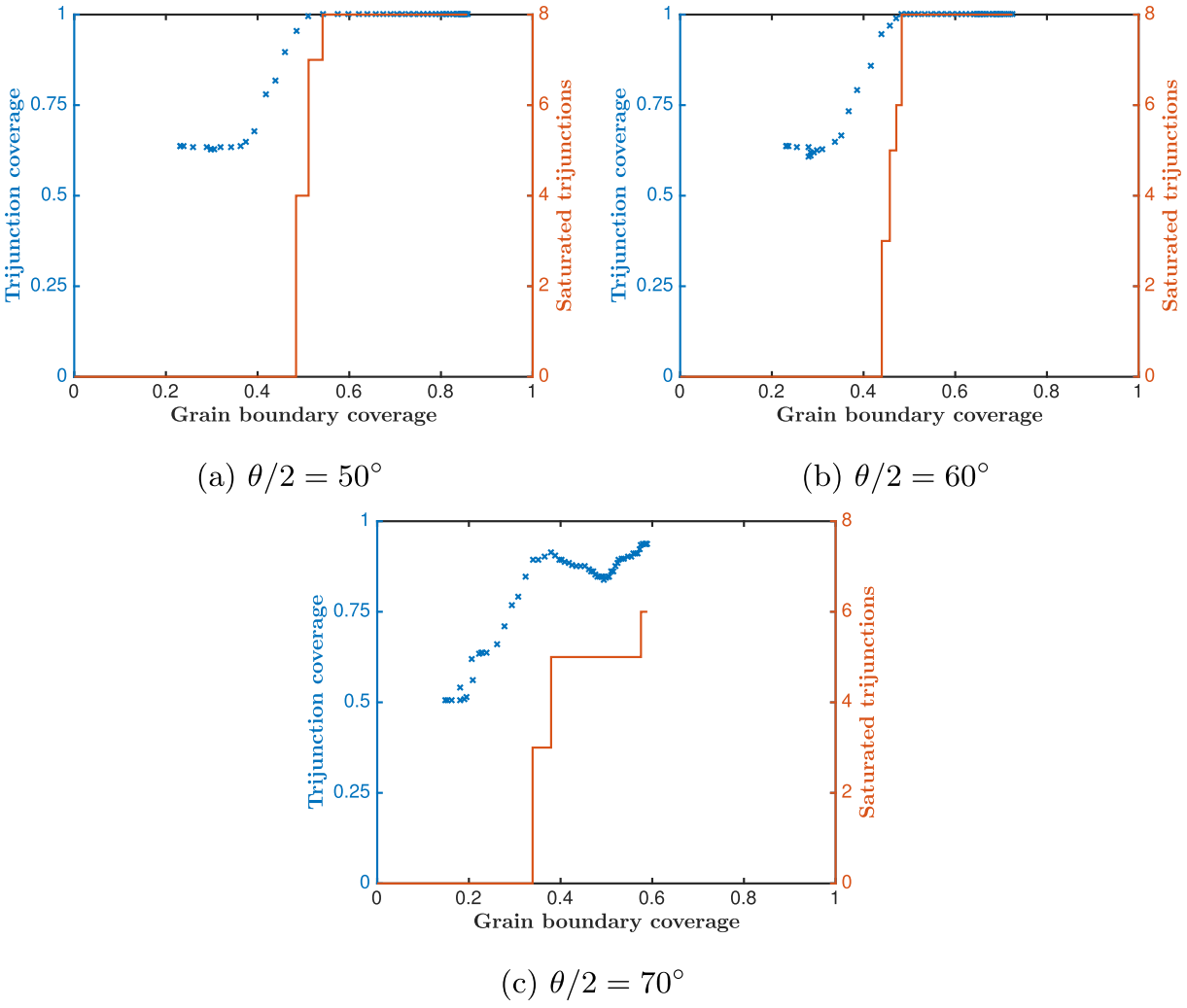


Fig. 8. Evolution of fractional triple junction coverage (X_{TJ}) and number of saturated triple junctions (n_s) as a function of grain boundary coverage by fission gas bubbles, for varying semi-dihedral angle $\theta/2$, with $s_v^0 = 10s_g^0$ and $N_{TJ} = 7.14/\mu m$. As the semi-dihedral angle increases, the triple junctions become saturated earlier with respect to grain boundary coverage. However, for larger $\theta/2$, not all triple junctions become fully saturated.

However, from Fig. 8c, it is seen that for $\theta/2 = 70°$, only six of the eight triple junctions become saturated. This was observed even as simulation time was increased over 1000 days. This is because of the change in morphology of the bubbles, as shown in Fig. 9. In the same way that the grain boundary bubbles do not spread out from the initial conditions as much when semi-dihedral angle increases, triple junction bubbles also do not spread out from the initial conditions as much along the triple junctions. The triple junction bubbles may instead preferentially connect with neighboring grain boundary bubbles, as seen in the red-circled triple junction bubbles in Fig. 9.

4. Discussion

An important question to consider is how these simulations relate to the fission gas release criteria used in engineering-scale fuel performance codes. One commonly used criterion is that gas release occurs from a local volume element as soon as $X_{GB} = 0.5$ at that element [22,23]. This criteria is based on observations of X_{GB} from experiment [14], and is consistent with the trends observed by Millett et al. [16], who found with phase-field simulations that the grain boundary bubbles became percolated in the range $0.4 < X_{GB} < 0.6$. For some of the cases considered here, the triple junctions become saturated at approximately the same range of grain boundary coverage, $0.4 < X_{GB} < 0.6$. However, under the conditions of high initial triple junction bubble density and large semi-dihedral angle, some triple junctions become saturated significantly before $X_{GB} = 0.5$, beginning as low as $X_{GB} = 0.34$.

Engineering-scale fuel performance codes do not commonly consider the need for a percolated pathway from the local volume element to a free surface to allow release to occur. However, Millett et al. found that accounting for the need for a percolated pathway to a free surface to allow for gas release may significantly reduce the gas release rate [18]. That work used 2D grain boundary percolation simulations, and thus did not consider the possibility of early formation of tunnels along triple junctions. Since the present phase-field simulations show that in some cases, triple junctions do not saturate significantly before grain boundaries become percolated, it is still believed that the need for a percolated pathway for gas release is an important factor which should be accounted for in engineering-scale fission gas release models. To confirm this, future work should attempt to determine the correct density of triple junction bubbles in the initial conditions for different fuel characteristic and operating conditions, perhaps by coupling with an atomistic technique that can realistically model nucleation. Because of the strong effect of semi-dihedral angle and the fact that grain boundary bubbles with widely varying contact angles are observed in close proximity in experiment [32], large-scale polycrystalline microstructures with a variety of bubble contact angles should also be conducted. Finally, the percolation threshold for triple junctions in UO_2

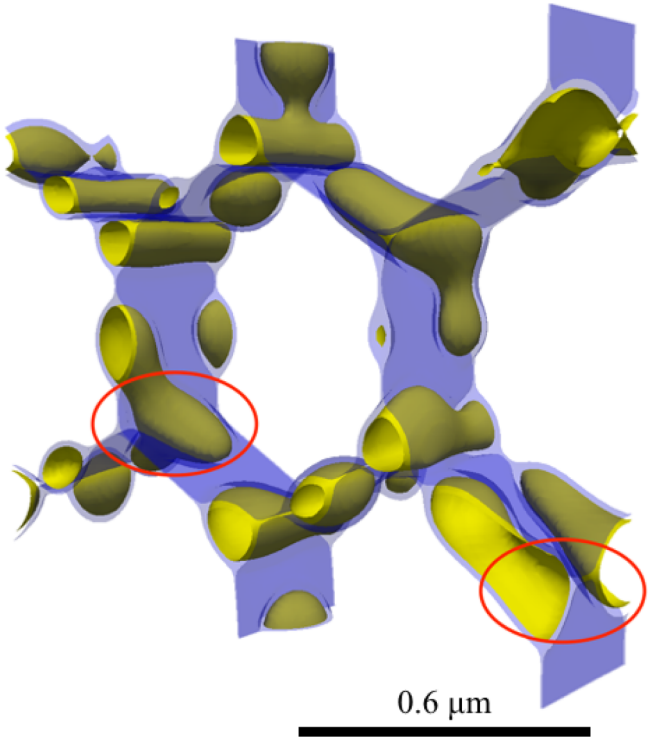


Fig. 9. Microstructure at 564 days for higher semi-dihedral angle $\theta/2 = 70°$. At this time grain boundary coverage is 0.46. $s_v^0 = 10 s_g^0$, and $L_z = 280$ nm, corresponding to initial $N_{TJ} = 7.14/\mu m$. Two of the eight triple junctions (circled in red) are not saturated, and do not become saturated even after >1000 days. This is in contrast to lower semi-dihedral angles, where all triple junctions become saturated.

should be quantified for more realistic grain structures. The correlations in Section 3.3 can be used with the percolation threshold in engineering-scale calculations to determine whether the triple junction network is locally percolated based on local grain boundary coverage, which can be used to determine whether a percolated path to a free surface exists for fission gas release.

5. Conclusions

Fission gas release has potentially negative consequences for fuel performance in light water reactors. To gain an improved understanding of the processes leading to fission gas release, a phase-field model has been used to simulate the growth and interconnection of grain boundary and triple junction bubbles in UO_2 fuel. The phase-field model is based on a grand-potential functional and allows simulation of multiple grains of fuel and the bubble phase. The solid phase free energies are based on an ideal solution model, while the gas phase free energies are based on the free energy of a van der Waals gas. Parabolic approximations are used for the bulk free energy of each phase. Source terms are used to include the production of fission gas atoms and the net production of vacancies.

The model was used to simulate the growth of grain boundary and triple junction bubbles in a hexagonally periodic geometry. Because engineering-scale fuel performance models commonly use grain boundary coverage as a criterion for fission gas release, the fraction of triple junctions covered and the number of triple junctions were plotted as a function of grain boundary coverage. When the density of triple junction bubbles in the initial conditions was increased, as expected, the triple junctions saturated earlier. The effect of vacancy source strength was found to have a relatively small effect on the value of X_{GB} at which complete triple junction saturation occurred. When the semi-dihedral angle of the bubbles was increased, saturation happened earlier with respect to grain boundary coverage, but not all triple junctions became fully saturated.

Ultimately, the objective of this work is to inform the fission gas release model used in engineering-scale fuel performance codes such as Idaho National Laboratory's BISON. Currently, BISON and some other codes track X_{GB}, and allows all gas within a local volume element to be released when $X_{GB} > 0.5$. Past work (where triple junctions were not considered) showed that accounting for the need for a percolated pathway to a free surface could significantly reduce the rate of fission gas release. The present work thus supports the previous work's conclusion that the need for a percolated pathway to a free surface should be included in the fission gas release model. Further work should focus on determining more physical nucleation densities for triple junction bubbles, and on larger simulations that include more grains and bubbles with varying semi-dihedral angles.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to legal reasons.

Acknowledgments

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